



GALERIE — Intelligent Multi-Vendor E-Commerce Marketplace

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Abstract

- GALERIE is a multi-vendor e-commerce marketplace that connects customers, vendors and administrators through a unified platform. The system uses React, Spring Boot, MySQL, Google OAuth2 and REST APIs, with an intelligence layer for vendor trust, demand analysis, inventory risk and review integrity.
- The platform aims to provide secure authentication, real product discovery, vendor management, intelligent marketplace analytics and a seamless digital shopping experience.

Tools & Technologies

- Front-End:** React, Vite, JavaScript, HTML, CSS
- Back-End:** Spring Boot, Java 21, REST APIs
- Database:** MySQL
- Authentication:** Google OAuth2 / Spring Security
- Payment:** Algorand Testnet
- Tools:** Spring Tool Suite, VS Code, MySQL, Postman, Git & GitHub

Existing vs Proposed System

Feature	Existing System	Proposed GALERIE
User Authentication	Basic / platform-dependent	Google OAuth2 authentication
Vendor Management	Limited vendor visibility	Dedicated Vendor Studio
Product Management	Basic product listing	Product + inventory management
Analytics	Limited marketplace insights	Demand, trust, inventory and vendor intelligence
Reviews	Basic ratings/reviews	Review integrity analysis
Payments	Conventional payment methods	Algorand Testnet-based digital payment
User Roles	Limited separation	Customer, Vendor and Admin roles
Intelligence	Mostly transactional	Data-driven intelligence + Gemini AI explanations

Feasibility Study

- Technical:** The system uses widely available technologies including React, Spring Boot, Java, MySQL, REST APIs, Google OAuth2 and AI integration.
- Economic:** The project uses open-source development tools and free/test environments wherever possible, reducing implementation cost.
- Operational:** Role-based interfaces allow customers, vendors and administrators to access functionality relevant to their responsibilities.
- Schedule:** The system is being developed incrementally during the semester, with separate implementation of authentication, marketplace modules, intelligence and integrations.

Results & Testing

- Successfully implemented Spring Boot REST backend.
- MySQL database and marketplace entities created.
- Real product dataset imported into the marketplace.
- Google OAuth2 authentication successfully implemented.
- Customer, Vendor and Admin roles supported.
- Product and category APIs tested successfully.
- Vendor profiles and inventory management supported.
- REST APIs can be tested using Postman.
- Intelligence modules are designed to derive insights from marketplace data.

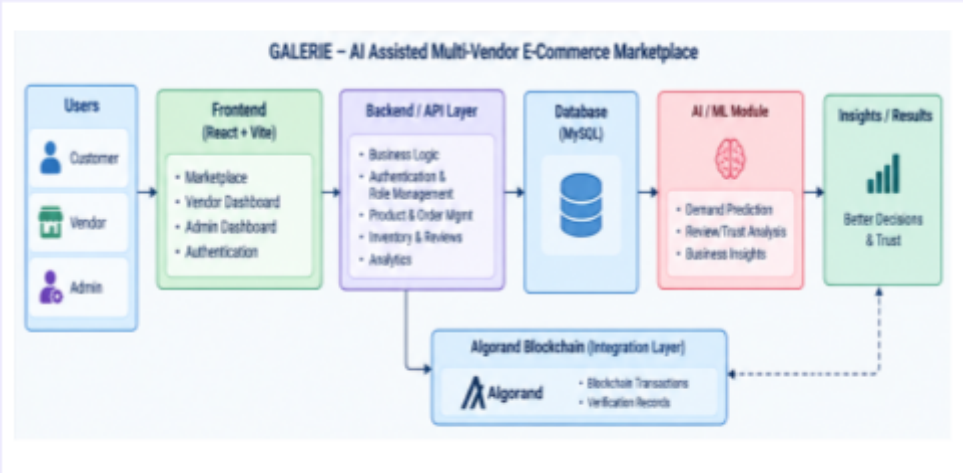
Conclusion

- GALERIE provides a multi-vendor e-commerce platform combining marketplace functionality with data-driven intelligence. The system integrates secure Google authentication, product and inventory management, orders, reviews, vendor operations and analytics through a modular Spring Boot and React architecture.
- The proposed system extends conventional e-commerce by adding intelligence capabilities for understanding demand, vendor trust, inventory risk and review integrity.

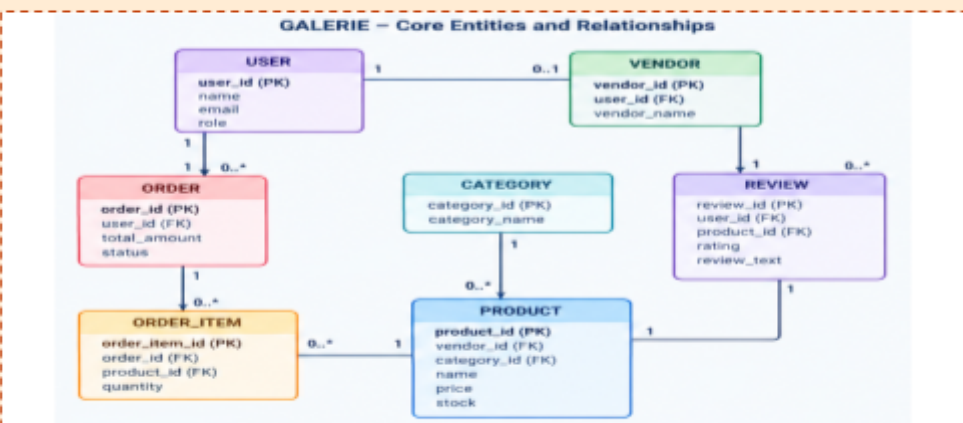
Problem Statement & Objectives

- Problem:** Traditional e-commerce platforms provide shopping and vendor management but offer limited transparency into vendor reliability, product demand, inventory risk and review authenticity.
- Objective 1:** Develop a secure multi-vendor marketplace supporting Customer, Vendor and Admin roles.
- Objective 2:** Implement product, inventory, cart, order, payment and review management using a Spring Boot REST backend and MySQL database.
- Objective 3:** Provide an intelligence layer for Trust & Reputation, Demand Intelligence, Review Integrity, Inventory Risk and Vendor Analytics.
- Scope:** The system covers customers, vendors and administrators with modules for authentication, product discovery, shopping cart, orders, reviews, vendor management, analytics and intelligent insights.

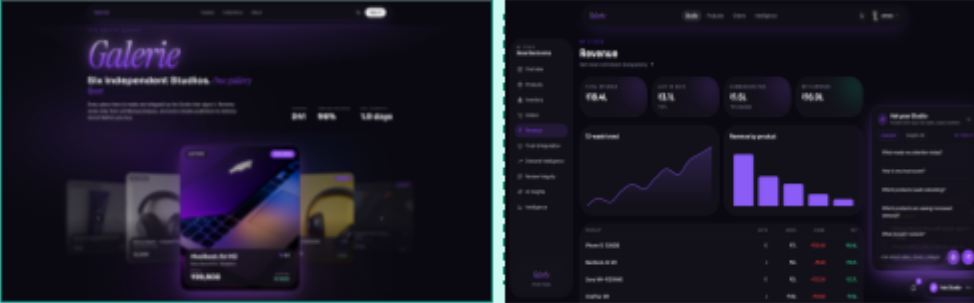
System Architecture



ER Diagram & Database Design



Implementation Screenshots



Future Scope

- Advanced demand forecasting using larger historical datasets.
- Automated vendor recommendations and personalized customer experiences.
- Expanded Gemini-powered AI assistance and natural-language analytics.
- Production-grade Algorand payment integration.
- Cloud deployment and scalable distributed services.
- Advanced fraud and review anomaly detection.

